

CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

CIA– III
ML FOR SOCIAL GOOD CHALLENGE
DOMAIN: Earth Environment
Forest-Fire Risk Prediction

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1. Overview

Forest fires can cause serious environmental damage. This project uses machine learning to predict forest-fire risk from weather and fire-weather information. The aim is to support forest monitoring teams in identifying conditions associated with higher fire risk.

The project uses ensemble machine learning and compares a Logistic Regression baseline with Random Forest, XGBoost and a Stacking Ensemble. The final system also includes SHAP-based explainability and a live synthetic prediction.

Prediction target: 1 = fire occurrence (area > 0); 0 = no fire occurrence (area = 0).

Intended beneficiaries: forest departments and environmental/fire-monitoring teams.

2. Dataset

Dataset: UCI Forest Fires Dataset.

Source: UCI Machine Learning Repository – Forest Fires.

The dataset contains 517 observations. Each observation represents an environmental/forest-fire record from the Montesinho region of Portugal.

Important variables include temperature, relative humidity, wind, rain and Fire Weather Index variables such as FFMCI, DMC, DC and ISI.

Target variable: burned area (area), converted into a binary fire-risk target for classification.

Dataset source: <https://archive.ics.uci.edu/dataset/162/forest%2Bfires>

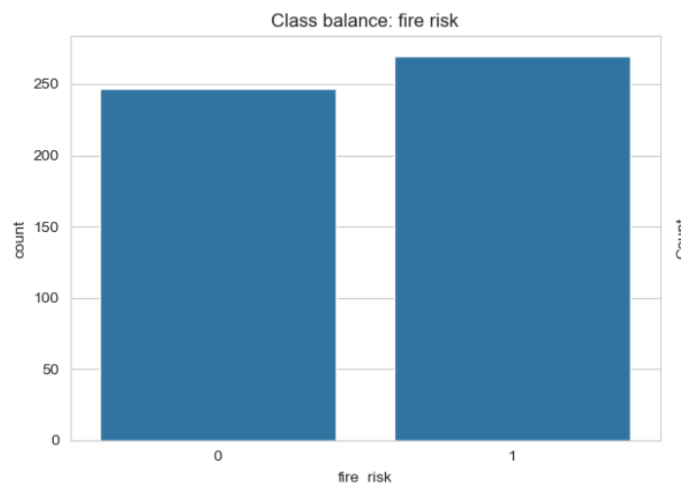


Figure 1. Target Class Distribution

3. Preprocessing

The dataset was checked for missing values, duplicate records, invalid values, data types and class distribution.

Missing values: No missing values were identified.

Duplicates: Four duplicate rows were identified and should be removed before the final modelling run.

Categorical variables such as month and day are encoded for modelling.

The original area variable is not used as an input feature because it is used to create the target. This prevents target leakage.

A stratified train/test split is used, and cross-validation is used during model development. The final test set is kept untouched until model comparison.

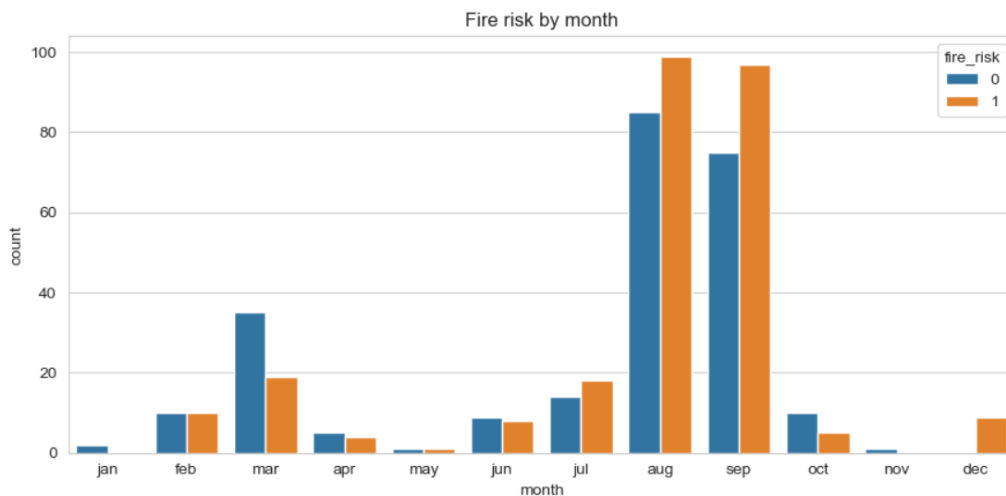


Figure 2. EDA / Data Analysis

4. Feature Engineering

Domain-informed features are created from the available weather and fire-weather information. These features represent conditions such as dryness, temperature, wind and seasonal effects that can be related to fire occurrence.

The engineered features are used together with the original valid predictors in the modelling pipeline.

5. Machine Learning Models

Four models are evaluated:

1. Logistic Regression – baseline model.
2. Random Forest – bagging ensemble.
3. XGBoost – boosting ensemble.
4. Stacking Ensemble – Random Forest and XGBoost as base learners with Logistic Regression as the meta-learner.

Models are developed using cross-validation and compared using the same untouched test set.

6. Results

Model	ROC-AUC	F1	Precision	Recall
Logistic Regression	0.600	0.610	0.627	0.593
Random Forest	0.631	0.626	0.590	0.667
XGBoost	0.674	0.629	0.647	0.611
Stacking	0.677	0.720	0.634	0.833

The Stacking Ensemble gives the best overall result, with the highest ROC-AUC and F1 score. Its recall is 0.833 compared with 0.593 for the Logistic Regression baseline. This is useful for fire-risk screening because missing a genuine high-risk case can be more serious than producing an additional warning.

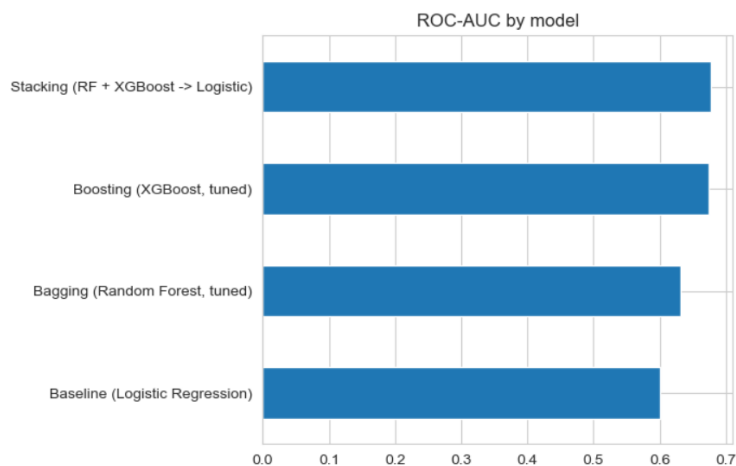


Fig 3. Model Performance Comparison

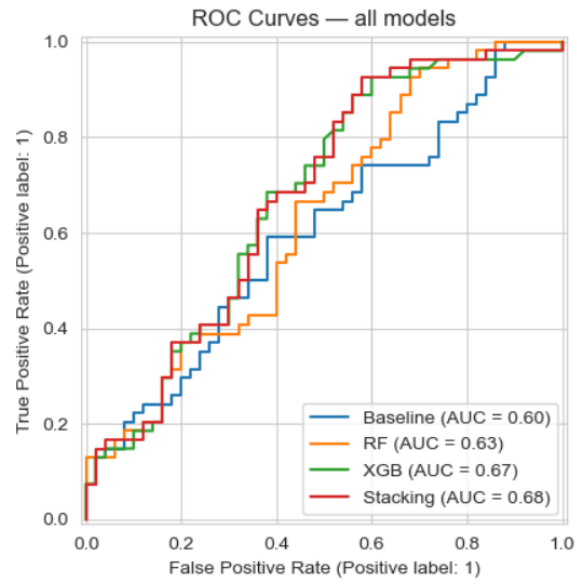


Fig 4. ROC Curve

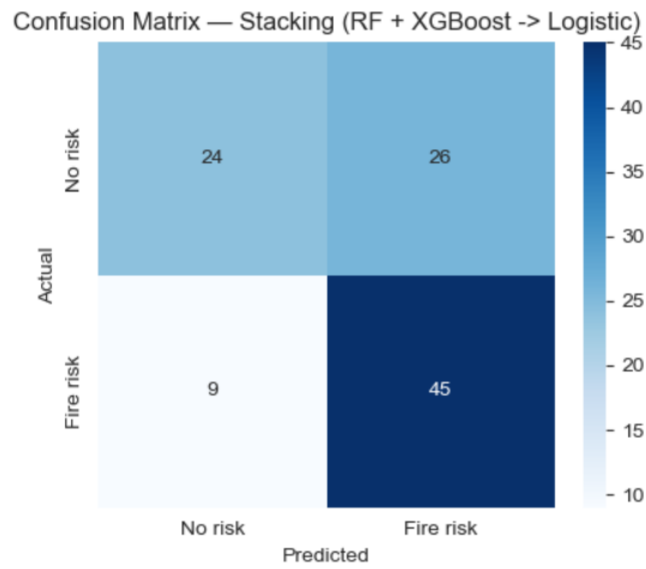


Fig. Confusion Matrix

7. SHAP Explainability

7.1 Global Explanation

Global SHAP analysis is used to identify the features that have the greatest influence on model predictions across the dataset. The most influential features are interpreted in terms of their environmental meaning.

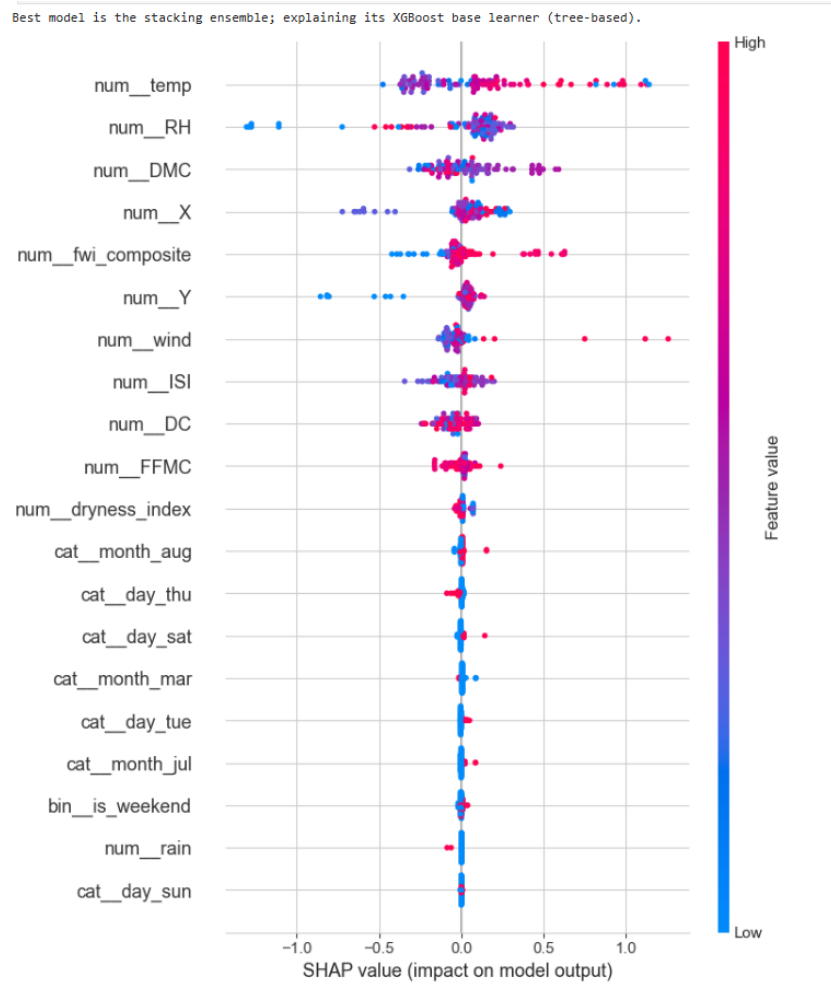


Figure 7. Global SHAP Explanation

7.2 Local Explanation

Local SHAP analysis explains one individual prediction by showing which features contributed to a higher or lower predicted fire risk.

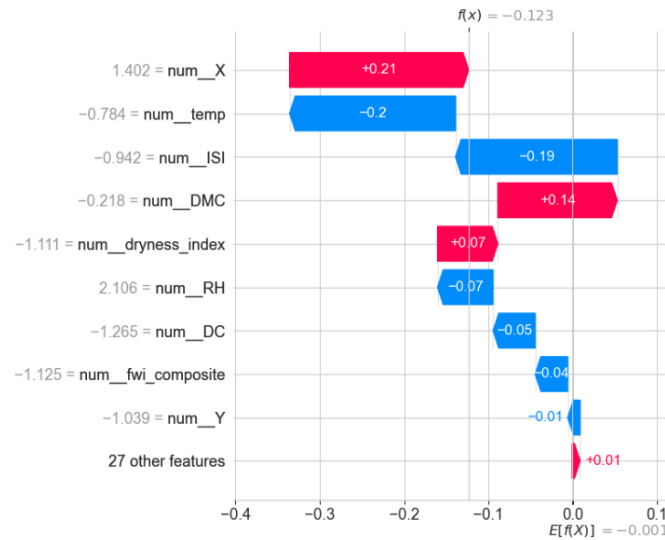


Figure 8. Local SHAP Explanation

8. Live Prediction

A realistic synthetic weather record is passed through the trained model to demonstrate the prediction process.

The demonstrated output is a fire-risk probability of **0.64**.

9. Ethics and Limitations

The dataset contains only 517 observations and is geographically specific to Portugal, so the model may not generalise to other regions. False negatives may be costly because a genuine fire-risk case could be missed, while false positives may lead to unnecessary monitoring. Model predictions contain uncertainty and should not be treated as guarantees. Human experts should remain involved in any real-world decision. The project uses a public environmental dataset and does not use personally identifiable information.

10. Conclusion

This project demonstrates an end-to-end ensemble machine-learning approach for forest-fire risk prediction. The Stacking Ensemble achieved the strongest reported performance, while SHAP provides global and local explanations. The live synthetic prediction demonstrates the use of the trained model. Further validation with larger and geographically diverse datasets would be required before real-world deployment.

